ARYA MADDEL - Machine Learning Engineer

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OBJECTIVE - ML engineer and researcher focused on cyber-physical systems - embedding physics-informed and efficient ML into robotics, aerospace, and real-time embedded hardware. IEEE-published (ETFI 2026) with production MLOps experience; driven to build intelligent systems that sense, decide, and act reliably on cheap, resource-constrained platforms. Target: Research Engineer / ML Architect in robotics, aerospace, or embedded systems.

EXPERIENCE

- **ERIC Robotics: Machine Learning and Robotics Intern** | Nov 2026 - Present
  Developing computer-vision capabilities and integrating machine-learning components into robotics workflows.
- **In2Peta Services Private Limited: Machine Learning Intern** | Feb 2026 - Nov 2026
  Designed and deployed ML pipelines using Python and Docker, implementing CI/CD-based MLOps workflows for automated training, versioning, and deployment.
- **Aerial Mappers & Surveyors LLP: Project Lead** | Jan 2024 - Mar 2024
  Developed an autonomous survey/delivery drone system on the Pixhawk open-source flight controller with variable payload capability; validated in the field over 7.5 km total coverage at a 350 g payload.
- **Sujalam Chemicals: Project Lead / Full-Stack Developer** | May 2023 - Aug 2023
  Built a cross-platform inventory management app (Android/iOS) using React Native, Express, SQLite, Ngrok; handled UI, backend, and database.

PROJECTS

- **Autonomous Drone** - Winner of the Nirmitee Project Presentation. Built a Pixhawk-based drone platform with waypoint navigation, payload handling (up to 350 g), and flight parameter tuning for delivery use cases; field-tested over 7.5 km total coverage.
- **OpenAnim** - Multi-tool LLM agent platform for programmatic animation. Routes natural language prompts to deterministic renderers (Manim for math animation, Kroki for diagrams) via an LLM orchestrator with tool-calling, SSE streaming, web search (Tavily), RAG over documentation, and content-addressed caching. Designed for extensibility - evaluated Remotion integration for browser-based video rendering.
- **DIQA** - No-Reference Image Quality Assessment (NR-IQA) framework with a unified interface and dynamic routing mechanism for model selection (Published at IEEE ETFI 2026, Maddel et al.). GitHub
- **IN2PETA Coding Plan & LLM Gateway** - Designed and built the SaaS coding platform from scratch (subscription tiers, usage-based billing, model catalog, API key management) and integrated a LiteLLM gateway with 11+ models across Nebius and Fireworks - smart routing, usage tracking, budget enforcement. Built the Fastify API and Next.js frontend.

SKILLS

- **Embedded & Edge ML:** STM32 (F411/H7), Pixhawk/PX4 flight controllers, TinyML deployment, C/C++
- **ML & AI:** PyTorch, physics-informed neural networks (PINNs), sensor fusion, deep learning, computer vision / image processing, scikit-learn, LightGBM
- **MLOps & Software:** Python, Docker, CI/CD, Linux, Git, MLOps pipelines, LLM tooling & orchestration (LiteLLM)
- **Web:** Next.js, React Native, Fastify/Express, SQLite

EDUCATION

- Bachelor of Technology in Computer Science and Engineering | DES Pune University | CGPA: 8.76 | Aug 2024 - Jun 2027
- Diploma in Computer Engineering | Sou. Venutai Chavan Polytechnic | 86.4% | Sep 2021 - Jun 2024
- ICSE Board | Hutchings High School | 90.67 % | 2011-2021

PUBLICATIONS

Maddel et al., "**Adaptive Hybrid Architecture for NRIQA**," in *Proceedings of the 2026 International Conference on Emerging Technologies and Future Innovations (ETFI)*, Pune, India, 2026, pp. 1-7, doi: 10.1109/ETFI68128.2026.11484766. Proposes a dynamic routing pipeline that predicts and executes the optimal NR-IQA model using low-cost perceptual features and a gradient-boosted classifier, reducing latency while maintaining performance comparable to deep models like HyperIQA.